

Human Trust and Perception of LLM-Generated Content

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Abstract

We present the first pre-registered, large-scale, controlled empirical study of human trust in LLM-generated content within African populations. Across $N=412$ participants from 12 African nations spanning four regions, we crossed **source** (human / GPT-4-Turbo), **disclosure** (shown / blind), and **domain** (news/science / legal) using 24 African-contextualised passages. Trust was measured with the 12-item Schaefer et al. (2016) HRI Trust Perception Scale, alongside credibility, information quality, and fact-checking intent.

Four findings stand out: **(i)** disclosing LLM authorship reduced trust by 34.3% ($M=3.08$ vs 4.63, Cohen's $d=1.45$); **(ii)** in the blind condition, participants could not distinguish LLM from human text at above-chance rates (53.8%, $p=.13$); **(iii)** the penalty was largest in the legal domain (stakes-sensitivity); **(iv)** AI literacy, internet-access type, and urban/rural status emerged as a significant moderator of the latter two without parallel in WEIRD-sample research.

Motivation: A WEIRD Blind Spot

Almost all empirical evidence on human trust in LLM-generated content has been gathered from Western, Educated, Industrialised, Rich, and Democratic populations (Henrich, Heine, & Norenzayan, 2010). Yet Africa, home to over 1.4 billion people speaking more than 2,000 languages, is the world's fastest-growing mobile internet market and is characterised by radically different information access conditions: predominantly 4G/5G mobile, news consumed via radio and WhatsApp rather than online publishers, and AI literacy levels that vary substantially across urban, peri-urban, and rural contexts.

If humans cannot reliably distinguish LLM-generated content from human-authored text, and if disclosure systematically shifts perception regardless of actual quality, then the deployment of LLMs at a continental scale could distort the African information ecosystem in ways that current NLP benchmarks, almost none of which include Sub-Saharan participants, cannot capture. This study supplies the missing evidence.

Research Questions and Hypotheses

- RQ1 - Source disclosure. Does explicitly disclosing that a text is LLM-generated reduce trust, credibility, and information quality ratings relative to human-authored text? *H1a*: disclosed LLM < disclosed the human. *H1b*: In the blind condition, participants do not distinguish LLM from human text above chance.
- RQ2 - Domain moderation. Does the trust gap vary across news, science, and legal content? *H2*: largest in legal, smallest in news (stake sensitivity).
- RQ3 - AI literacy moderation. Does individual AI literacy reduce the disclosure penalty? *H3*: higher AI literacy smaller penalty, via content-focused rather than source-based evaluation.
- RQ4 - African contextual moderators. Do internet-access type and urban/rural classification moderate the penalty? *H4a*: 2G/3G users show larger gaps than broadband users. *H4b*: rural participants show larger gaps than urban participants.

Method

Design. Pre-registered 2 (Source: Human / LLM) $\times$ 3 (Domain: News / Science / Legal) $\times$ 2 (Disclosure: Shown / Blind) mixed design. Source was between-participants; Domain and Disclosure were within-participants and counter-balanced.

Stimuli. 24 African-contextualised passages ($\approx$ 200 words; 12 matched human-LLM pairs, 4 per domain). Topics included the African Union Climate Summit, AfCFTA trade progress, Sahel drought food security, the H3Africa hypertension programme, Nile Basin water-stress projections, the African Court land-rights ruling, Kenya Data Protection Act enforcement, and the South African Mining Charter constitutional ruling. Human texts were drawn from *AllAfrica.com*, *Daily Nation*, *Punch Nigeria*, the H3Africa Consortium, and publicly available African court rulings. LLM texts were produced with **gpt-4-turbo** (temperature 0.7, **max_tokens** = 350) on matched topics and reading levels.

Equivalence pilot ($n=48$). Human and LLM passages matched on Flesch-Kincaid grade (13.0 vs 12.8, $p=.44$) and on factual accuracy as scored blindly by three domain experts (4.82/5.0 vs 4.77/5.0, $p=.27$). **Measures.** Trust: 12-item Schaefer et al. (2016) HRI scale, three sub-scales (competence, integrity, benevolence), 7-point Likert; Cronbach's $\alpha \in [.81, .89]$. Perceived credibility: 3-item Web Credibility Scale (Fogg et al., 2003). Information quality: 4 items (clear, well-organised, comprehensive, accurate). Fact-checking intent: a single behavioural item, How likely are you to verify the information in this text before using it? (1–7). AI literacy: 25-item Carolus et al. (2023) MAILS scale, sample $M=3.36$, $SD=0.73$. **Procedure.** Informed consent $\rightarrow$ demographics + AI literacy $\rightarrow$ random Source assignment $\rightarrow$ three passages (one per domain, counter-balanced disclosure), $\rightarrow$ trust/credibility/quality/fact-check items, $\rightarrow$ source guess (blind condition) $\rightarrow$ debrief. Manipulation check: 93.8% Participants in the disclosed condition correctly recalled the stated authorship.

Participants ($N = 412$)

Recruited via Prolific Academic's Africa panel ($N=460$ enrolled; 48 excluded for failed attention checks or incomplete submissions).

Region	Countries (n)	Total
West Africa	Nigeria (88), Ghana (37), Senegal (12)	137
East Africa	Kenya (60), Ethiopia (40), Tanzania (30), Uganda (21), Rwanda (12)	163
Southern Africa	South Africa (73), Zambia (17), Zimbabwe (12)	102
Central Africa	Cameroon (10)	10
Total		412

The sample was heterogeneous across the dimensions theorised to shape AI trust: 50.2% urban / 23.1% peri-urban / 26.7% rural; 41.5% mobile 4G, 34.2% mobile 2G/3G, 17.0% broadband, 7.3% no regular internet; 42 language communities represented (English 93, Swahili 48, Hausa 27, Igbo 21, Zulu 19, Yoruba 18, Twi/Akan 13, Amharic 12, plus 34 further languages). Sample AI literacy ($M=3.36$) was meaningfully lower and more variable than typical US baselines ($M>3.8$), giving the design strong power to detect moderation effects.

Limitations & Future Work

The 12 country sample is not representative of the continent's full diversity: North Africa and the Horn of Africa are not included, and Prolific recruitment skews towards younger, better-educated, more connected adults. All stimuli were administered in English; only 22.6% of participants reported English as their primary language. Because the central comparison is within-participant and within-language, any second-language reading penalty applies equally across labelling conditions and cannot account for the 34.3% disclosure gap or its consistency across countries. External validity for LLM content delivered in African languages is not established here; a Swahili / Hausa/ Yoruba / Zulu / Amharic replication is planned.

Lexical Equivalence and Detection (H1b)

A post-hoc lexical comparison of all 24 passages found no differences in length, type-token ratio, MTLD, word length, or Flesch-Kincaid grade (all $p > .05$). The only reliable surface difference was a slightly longer mean sentence length in LLM passages (32.4vs 28.2 words; $t(22)=-2.60$, $p=.017$, $d=-1.06$) a cue insufficient to support above-chance authorship judgements: blind detection accuracy was 53.8% (95% CI [49.1%, 58.5%], $\chi^2(1)=2.31$, $p=.13$). *The 34.3% disclosure penalty, therefore, cannot be attributed to detectable stylistic cues; it is a label effect.*

RQ1 - The Source-Disclosure Penalty

A 2 (Source) $\times$ 2 (Disclosure) mixed ANOVA on overall trust scores revealed a significant **Source** $\times$ **Disclosure** interaction, $F(1, 410) = 84.3$, $p < .001$, $\eta^2 = .17$. The gap was confined to the disclosed condition (Table 1). Identical patterns held for perceived credibility (disclosed LLM $M=3.31$ vs human $M=4.78$, $p < .001$) and for fact-checking intent, where the disclosed-LLM condition was elevated above all others ($M=5.24$ vs ≤ 4.50), indicating that disclosure shifts *behaviour*, not only attitudes.

Table 1. Mean trust and outcome scores by Source $\times$ Disclosure (SD in parentheses; 7-point Likert).

Condition	Trust	Credibility	Info Quality	Fact-Check
LLM (Disclosed)	3.08 (0.93)	3.31 (0.90)	3.52 (0.96)	5.24 (1.04)
Human (Disclosed)	4.63 (0.86)	4.78 (0.81)	4.89 (0.84)	3.51 (1.09)
LLM (Blind)	4.47 (0.89)	4.43 (0.87)	4.55 (0.92)	4.38 (1.06)
Human (Blind)	4.55 (0.91)	4.51 (0.85)	4.61 (0.89)	4.30 (1.01)

Disclosed contrast: $t(410) = 16.7$, $p < .001$, $d = 1.45$ · Blind contrast: $t(410) = 0.88$, $p = .38$, $d = 0.09$.

RQ2 - Domain Sensitivity: A Stakes Heuristic

A Source $\times$ Domain ANOVA restricted to the disclosed condition yielded a significant interaction, $F(2, 1230)=24.1$, $p < .001$, $\eta^2=.04$. The trust gap was largest in the legal domain ($\Delta=1.62$), intermediate in science ($\Delta=1.28$), and smallest in news ($\Delta=1.08$). The legal vs news difference was significant ($p < .001$, $d=0.44$); legal vs science was marginal ($p=.06$). The pattern is consistent with a *stakes-sensitivity* account: when the consequences of acting on incorrect information are highest, readers apply stronger source-based discounting. The sizable science gap likely reflects the visibility of African research institutions (H3Africa, Nile Basin Initiative) in our stimuli, which may activate stronger expectations of institutional credibility.

Table 2. Trust gap (Human – LLM) by content domain in the disclosed condition. All $p < .001$ Bonferroni-corrected.

Domain	LLM trust	Human trust	Gap (Δ)	d
News	3.30	4.38	1.08	1.21
Science	3.02	4.30	1.28	1.44
Legal	2.83	4.45	1.62	1.78
Overall	3.08	4.63	1.55	1.45

RQ3 & RQ4 - Who Discounts AI Most?

A moderated regression on disclosed-condition trust scores (Source dummy $\times$ mean-centred AI literacy; covariates: domain, country) revealed a significant Source $\times$ AI Literacy interaction, $B=0.19$, $SE=0.04$, $\beta=.22$, $t(407)=4.87$, $p < .001$. Simple slopes: low AI-literacy readers (-1 SD) showed a 1.71-point penalty; high AI-literacy readers ($+1$ SD) showed a 0.89-point penalty a 48% **reduction**. Crucially, AI literacy did not predict trust ratings in the blind condition (all $p > .21$), confirming that literacy shapes responses to the *source label* rather than baseline evaluations of text quality.

Extending the model with internet-access and urban/rural classification revealed two additional moderators with no parallel in WEIRD-sample research (Table 3). The internet-access effect was partially mediated by AI literacy (Sobel $z=3.21$, $p=.001$; 41% mediation), reflecting lower AI-literacy scores among 2G/3G users ($M=3.22$ vs $M=3.52$ for 4G; $t(399)=4.12$, $p < .001$).

Table 3. Disclosed-condition trust gap (Human – LLM) by African contextual moderator.

Moderator	Trust gap	Test
<i>Internet access</i>		$F(2, 409)=8.43^{***}$
Broadband / WiFi	1.09	
Mobile 4G	1.31	
Mobile 2G/3G	1.88	
<i>Urban / rural</i>		$F(2, 409)=6.84^{**}$
Urban	1.42	
Peri-urban	1.78	
Rural	2.04	
<i>Primary news platform</i>		
Online news	1.24	
WhatsApp	1.78	
Radio	1.92	
<i>AI literacy</i> ($- / + 1$ SD)	1.71 $\rightarrow$ 0.89	$\beta=.22^{***}$

$^{**}p < .01$, $^{***}p < .001$.

Discussion

Disclosure is not neutral. The 34.3% source-disclosure penalty replicates and amplifies effect sizes reported in US and European studies (Huschens et al., 2023; Baek, Kim, & Kim, 2026; Akpinar et al., 2026), despite participants being unable to detect LLM content above chance, confirming that trust responses are driven primarily by source heuristics rather than textual quality. Mandatory AI-generated labels, increasingly written into AI policy globally, will therefore measurably suppress trust even when the content is indistinguishable in quality from human writing.

Two novel moderators specific to the African digital landscape. 2G/3G users showed trust gaps $\sim$ 44% larger than broadband users; rural participants showed gaps 44% larger than urban participants. These patterns sit at the intersection of digital-infrastructure inequity and AI-trust calibration – a class of moderator absent from desktop-centric, broadband-saturated Western samples. They imply that transparency and disclosure frameworks must account for connectivity and urban/rural variation, that Human-annotation studies should report AI literacy, connectivity, and geographic origin, and that mobile-first, low-bandwidth, multilingual AI literacy resources are an urgent priority for the Global